

Research methodology: Every component below was individually searched and verified against current vendor listings on July 04, 2026. URLs are listed for direct verification. **V** = price confirmed at the named URL · **E** = commodity item; market estimate from multiple sources · **O** = operator's own documented purchase. Where prices differ materially from earlier planning figures, the Notes column explains why.

COMPONENT	PRICE	ST	SOURCE URL	VERIFIED PRICE DETAIL	NOTES
FIELDKOMMS SERVER (192.168.50.1)					
Raspberry Pi 5 — 16 GB	\$305.00	V	raspberrypi.com/products/raspberry-pi-5	\$305.00 — official Raspberry Pi product page	3rd memory-shortage hike April 1 2026 (was \$120 at launch). Forum table at forums.raspberrypi.com confirms \$305 current. Expect decline 2027+.
SunFounder Pironman 5-MAX	\$94.99	V	sunfounder.com · amazon.com · microcenter.com	\$94.99 — Amazon/Micro Center listing	Dual NVMe RAID 0/1, OLED, RGB tower cooler, HAT+ slot. Micro Center SKU 699476.
1 TB NVMe SSD (x2 for RAID 1)	\$180.00 ea	O	amazon.com (owner actual purchase, June 2026)	\$180.00 each — Jim's documented Amazon purchase	Market range verified at storagediskprices.com: \$110–218 (Samsung 990 Pro \$218). NAND up ~4× in 9 months per TrendForce. Budget \$180 is mid-range.
Raspberry Pi 27W USB-C PSU	\$14.00	E	raspberrypi.com · amazon.com	~\$12–14 street	Official Raspberry Pi 27W PSU. Commodity; stable price.
MicroSD 32 GB	\$10.00	E	amazon.com	~\$8–12 street	Boot media / recovery image. SanDisk or Samsung A1 class. Commodity.
44NET GATEWAY PI (192.168.50.2)					
Raspberry Pi 5 — 16 GB	\$305.00	V	raspberrypi.com/products/raspberry-pi-5	\$305.00 — same source as above	8 GB variant (\$175) acceptable for gateway-only role if budget requires.
Argon NEO 5 BRED M.2 case	\$35.00	E	argon40.com · amazon.com	~\$35–40 street	M.2 SATA expansion case for Pi 5. Not affected by memory shortage.
256 GB M.2 SATA SSD	\$45.00	E	amazon.com · newegg.com	~\$40–55 street (NAND shortage)	Was ~\$25 in 2025. NAND shortage affects SATA too; verify at order time.
Raspberry Pi 27W USB-C PSU	\$14.00	E	raspberrypi.com · amazon.com	~\$12–14 street	Commodity.
NETWORKING					
ASUS RT-BE58 Go Wi-Fi 7 router (x3)	\$119.00 ea	V	amazon.com/dp/B0FSPQJGF · shop.asus.com	\$119.00 street — Tom's Hardware review Feb 2026 confirms price; MSRP \$159.99 at shop.asus.com	1 primary router + 2 AiMesh nodes. Wi-Fi 7, 2.5G WAN, USB WAN (cellular tethering), USB-C powered.
UniFi Switch Lite 16 PoE	\$199.00	V	store.ui.com/us/en/products/usw-lite-16-poe	\$199.00 — official Ubiquiti store, USW-Lite-16-POE	16× GbE, 8× PoE+ (802.3at), 45W PoE budget, fanless. Verified direct from Ubiquiti store.
CAT6 patch cables (x10 assorted)	\$30.00	E	amazon.com · monoprice.com	~\$3/ea street	Commodity bundle 1–6 ft lengths.
WAN — CELLULAR (INSTYCONNECT)					
InstyConnect Drum system	\$375.00	O	instyconnect.com/order	Owner documented purchase price	Complete 5G X62 POE + Drum antenna system. Drum antenna released early 2026 per rvmobileinternet.com review. Builds via order wizard at instyconnect.com/order — pricing varies by configuration. Verify before purchase.
InstyConnect Switchblade	\$425.00	O	instyconnect.com (owner purchase)	Owner documented purchase price	Directional folding 4× LDAP antenna array add-on. Reviewed at rvmobileinternet.com. Pricing via order wizard; not listed as standalone SKU on main page.
Spare outdoor PoE cable	\$20.00	E	instyconnect.com · amazon.com	~\$20 street	Weatherproof outdoor-rated shielded CAT6.
WAN — SATELLITE (STARLINK)					

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Starlink Standard Gen 3 kit	\$349.00	V	homedepot.com (SKU 329052560) · starlink.com	\$349.00 — Home Depot product page verified; starlink.com confirms hardware cost	PRICE DROP from \$499. Includes dish, Wi-Fi 6 router, cable, kickstand. Rental option: \$10/mo for Residential plan customers. Regional promo pricing as low as \$89 in some zones — check starlink.com for your address.
Starlink Ethernet adapter	\$25.00	V	starlink.com (accessories store)	\$25.00 — confirmed at starlink.com	Required to bypass Starlink router and connect dish to ASUS RT-BE58 Go.
USB 3.0 Gigabit Ethernet adapter	\$18.00	E	amazon.com	~\$15–20 street	ASIX AX88179 chipset recommended for compatibility. Commodity.
OPERATOR WORKSTATIONS (Pi 500+)					
Raspberry Pi 500+ Desktop Kit (×4)	\$430.00 ea	V	raspberrypi.com/products/raspberry-pi-500-plus · pishop.us	\$410 unit-only · \$430 Desktop Kit — raspberrypi.com official product page	Pi 500+ includes 16 GB RAM, 256 GB NVMe SSD, mechanical keyboard (Gateron KS-33 Blue). Desktop Kit adds mouse, 27W PSU, HDMI cable. Current price per April 2026 hike confirmed at forums.raspberrypi.com price table and Tom's Hardware Apr 2026. Originally \$200 at launch Sept 2025.
Raspberry Pi Monitor 15.6" (×4)	\$100.00 ea	V	raspberrypi.com/products/raspberry-pi-monitor	\$100.00 — official Raspberry Pi product page, price unchanged since Dec 2024 launch	15.6" 1080p IPS, USB-C powered from Pi USB port (60% brightness) or separate PSU. No DRAM content — unaffected by memory shortage.
HF STATION					
Icom IC-7300MK2	\$1,499.95	V	randl.com (R&L; Electronics) · gigaparts.com	\$1,499.95 — R&L; Electronics product page IC7300MK2	Next-gen IC-7300: HDMI output, USB-C dual COM + audio, LAN port (RS-BA1 remote, no PC needed), built-in CW decoder, RMDR ~105 dB, RX antenna IN/OUT. Watch hamradio.com/special_offers for Icom coupons — IC-7300 currently has \$300 instant coupon; MK2 coupon likely to follow.
Astron RS-35M-AP power supply (×2)	\$343.95 ea	V	theantennafarm.com (SKU 8941) · gigaparts.com · hamradio.com	\$343.95 each — The Antenna Farm product page verified · qty 2 = \$687.90	13.8 VDC linear regulated, 25 A continuous / 35 A peak, dual backlit meters, Anderson Powerpole outputs. Made in USA. Qty 2: primary for IC-7300MK2 + redundant/spare for field deployments.
LDG IT-100 autotuner	\$180.00	E	gigaparts.com · dxengineering.com/parts/ldg-it-100 · hamradio.com	~\$170–190 street across dealers	125 W Icom-specific autotuner. Powered and controlled from IC-7300 TUNER button via included IC-7K interface cable — no separate power needed. 2,000 memories. Exact price varies by dealer.
USB-A to USB-B cable	\$10.00	E	amazon.com · monoprice.com	~\$8–12 street	Radio to laptop for digital modes. Shielded recommended. Commodity.
Windows laptop	\$1,000.00	E	bestbuy.com · amazon.com · microsoft.com/surface	~\$900–1,100 mid-range laptop	Any Win 10/11 with USB-A port, Wi-Fi, SSD. Winlink Express + VARA HF + JS8Call. Budget \$1,000 for capable unit with fast storage and adequate USB ports. Many operators already own.
FIELD ANTENNA SYSTEM (CHAMELEON ANTENNA)					
CHA MPAS 2.0 modular antenna system	\$660.00	E	chameleonantenna.com/products/cha-mpas-modular-portable-antenna-system-2-0 · dxengineering.com/parts/cha-mpas-2-0	~\$650–670 from multiple dealers; DX Engineering confirms availability	1.8–54 MHz, 160m–6m. Includes HYBRID-MICRO matching unit, MIL 2.0 + MIL EXT 2.0 whips, spike mount, 73 ft + 25 ft wire, 50 ft coax with RFI choke, sling bag. 100 W SSB / 50 W CW. IMPORTANT: Chameleon USA250 Independence Day sale — 25% off select antennas through approx July 14, 2026 — could reduce MPAS 2.0 to ~\$495. Check chameleonantenna.com before purchase.

COMPONENT	PRICE	ST	SOURCE URL	VERIFIED PRICE DETAIL	NOTES
CHA URT1 universal remote tuner	\$389.95	V	randl.com (R&L; Electronics) · palomar-engineers.com (\$398.95)	\$389.95 — R&L; Electronics product page CHAURT1; \$398.95 at Palomar Engineers	Weatherproof feedpoint tuner. 1.8–54 MHz, 5–1,500 Ω matching range, 125 W SSB/CW, 16,000 memories. Power and control sent to tuner over existing coax via bias-T coupler — no radio-specific cable, works with any transceiver.
Chameleon antenna wire — 300 ft	\$300.00	E	chameleonantenna.com	~\$0.90–1.10/ft (25 ft and 73 ft spools)	Tinned copper with Kevlar PTFE jacket. For long wires, NVIS, dipoles, counterpoises, radial systems. Buy in spool quantities — verify current spool pricing at chameleonantenna.com at time of purchase.
Wire winders — heavy duty (×4)	\$15.00 ea	E	chameleonantenna.com · dxengineering.com	~\$12–18 ea	Tangle-free storage and rapid deployment of antenna wire in field conditions.
FACTOR HF MODEM					
SCS PXdragon DR-9400	\$2,099.00	V	seatech.systems/product/p4-dragon-dr-9400-packet-4-modem · dxengineering.com/parts/pxn-dr-9400-btwf	\$2,099 MSRP — confirmed at SeaTech Systems (authorized US dealer) and DX Engineering	Current-generation replacement for the discontinued DR-7800/7400 series (last sold Q1 2025). FACTOR-1/-2/-3/-4, Packet Radio 1k2 AFSK + 9k6 G3RUH, Robust Packet Radio. Quad-core ARM v8 @ 1.8 GHz running real-time Linux. 1-ppm TCXO for frequency stability. USB-C + Ethernet LAN standard (these were expensive options on DR-7800). Optional Wi-Fi/Bluetooth USB dongle and AES-256 encryption license available. Winlink, Airmail, RMS Express compatible. Manufactured in Germany; SeaTech handles US warranty repair without overseas shipping. Note: IC-7300MK2 connects via modem DIN audio port; USB-C to laptop for control.
IC-7300MK2 interface cable set	\$60.00	E	seatech.systems (custom cable orders) · dxengineering.com	~\$50–75 for audio + CI-V control cable set with ferrites	Two cables required: (1) audio cable from DR-9400 DIN port to IC-7300MK2 mic/speaker jacks; (2) CI-V control cable. Ferrite chokes on both recommended to reduce RF interference. Order from SeaTech or DX Engineering when purchasing the DR-9400 — specify IC-7300MK2 to get correct connectors.
VHF / UHF STATION					
Yaesu FTM-510DR	\$669.95	V	hamradio.com (HRO) — in stock, free shipping; gigaparts.com ; dxengineering.com	\$669.95 — Ham Radio Outlet verified in-stock price	144/430 MHz dual band C4FM digital + FM — 55W VHF / 50W UHF — built-in GPS + APRS (1200/9600 baud) — Super-DX weak-signal RF amp boost (both C4FM and FM) — 2.4" color touch display — detachable front panel — WiRES-X / Fusion digital voice. FTM-510DR ASP variant (\$749.95) includes ASP audio DSP board pre-installed — buy the ASP if budget allows.
Comet CA-2X4SRNMO dual band antenna	\$71.95	V	randl.com (R&L; Electronics) — CA2X4SRNMO · dxengineering.com/parts/cma-ca-2x4srnmo	\$71.95 — R&L; Electronics product page verified; same price PL-259 version (CA2X4SR)	Wideband 2m/70cm mobile vertical designed specifically for Search & Rescue and EMCOMM. Covers 140–160 / 435–465 MHz — includes Starcom VHF (155.xxx MHz) AND full ham bands on ONE antenna. 3.8/6.2 dBi gain. 150W. 40" with fold-over hinge. NMO connector. Note: CA-2X4SR (PL-259) also \$71.95 at R&L; if PL-259 preferred.
Comet CM-5NMO magnetic mount	\$35.00	E	amazon.com · dxengineering.com/parts/cma-cm-5nmo	~\$30–40 street	3.25" heavy-duty magnetic base, NMO connector, RG-58 coax tail. Mounts to vehicle roof or equipment case lid. Pairs directly with CA-2X4SRNMO.
RG-8X coax jumper	\$15.00	E	amazon.com · theantennafarm.com	~\$12–18 street	Short jumper to connect CM-5NMO coax tail to FTM-510DR antenna port. PL-259 both ends.

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Digirig Mobile rev 1.11	\$49.97	V	digirig.net/product/digirig-mobile	\$49.97 — verified directly at digirig.net product page	USB digital modes interface — combines audio codec, serial CAT and hardware PTT in one device — single USB-C connection to laptop — works Windows/Mac/Linux — enables FT8, JS8Call, VARA FM, SSTV, packet on the FTM-510DR — note: cables sold separately at digirig.net; order the Yaesu cable matching your radio. Note: Digirig on summer break until July 31, 2026 — orders ship automatically.
Mobilinkd TNC4	\$149.95	V	store.mobilinkd.com/products/mobilinkd-tnc4	\$149.95 — verified directly at Mobilinkd store page	1200/9600 baud KISS TNC — Bluetooth 4.2 dual-mode (iOS + Android compatible) + USB-C serial — AFSK/GFSK/4-FSK modulation — high dynamic range ADC/DAC — TCXO precision oscillator — 900mAh battery (48 hr continuous) — enables APRS in KISS mode for YAAC and Direwolf directly on FieldComms Pi — configured via Bluetooth app (iOS/Android). Cable sold separately at store.mobilinkd.com — order the Yaesu FTM mobile cable for FTM-510DR.
Digirig Yaesu FTM MiniDin10 cable	\$19.95	V	digirig.net/product/yaesu-ftm-cable (SKU: YAESU10MOBILE)	\$19.95 — verified at digirig.net product page; FTM-510DR explicitly listed in compatibility	10-pin MiniDin connector to Digirig audio socket (black jack). Audio + PTT. Enables VARA FM, Winlink, JS8Call, SSTV, APRS via Digirig + laptop. Works with all Digirig models. Note: Digirig on summer break until July 31, 2026.
Mobilinkd Yaesu FTM modem cable	\$24.95	V	store.mobilinkd.com/products/modem-cable-for-yaesu-ftm-mobile-radio	\$24.95 — verified directly at Mobilinkd store product page	10-pin MiniDIN connector, 1.5m length. Fits FTM-510DR (same MiniDIN-10 data connector as FTM-100/200/300/400/500 series). Connects TNC4 to radio for APRS KISS mode via Bluetooth or USB to FieldComms Pi.
USB-C data cable for Digirig	\$10.00	E	amazon.com	~\$8–12 street	Standard USB-C cable supporting both power AND data (not all USB-C cables support data). Not included with the Digirig unit — must be purchased separately. Connects Digirig to laptop USB port.
TRANSPORT CASES					
Harbor Freight Apache 4800 XL weatherproof case (x6)	\$59.99 ea	V	harborfreight.com (Item 64250 black; 56863-56866 colors) — confirmed \$59.99 list	\$59.99 list × 6 = \$359.94 — harbor Freight website verified; compare \$216.95 Pelican 1550	IP65 rated (dustproof + watertight). Interior 17-7/8 × 12-7/8 × 6-5/8 in. Pick-and-pull foam, purge valve, double-throw latches. Coupon code brings to \$49.99 (~\$300 total); Inside Track Club members pay \$39.99 (~\$240 total) — watch for coupons before purchase. Available in black, tan, yellow, orange, green. Suggested assignment: 1-Server, 2-Networking/WAN, 3-HF Radio, 4-VHF/UHF + Antenna, 5-Workstation cables/accessories, 6-Chameleon antenna system.
PRINTING & ACCESSORIES					
Brother MFC-L3780CDW color laser MFP	\$450.00	E	bestbuy.com · walmart.com · amazon.com/dp/B0CFCW8PGV	~\$430–500 street (Best Buy, Walmart, Amazon listings confirmed)	Successor to MFC-L3770CDW (discontinued). 31 ppm color laser-quality, duplex, Wi-Fi + GbE, ADF. Full Linux CUPS support. \$870 w/ 4-pack toner bundle at Walmart. IAP packages, ICS forms, operator cards.
LaCie Rugged 1TB USB-C (STFR1000800)	\$125.00	V	compsource.com (\$123.19, Apr 2026) · amazon.com/dp/B01N7BN0YF · bestbuy.com · bhphotovideo.com	\$123.19 — CompSource verified Apr 22, 2026; ~\$120–130 across dealers	Drop resistant 4 ft, crush resistant 1 ton, rain resistant. 130 MB/s USB-C 3.1 Gen 1. Includes USB-C + USB-C to USB-A adapter. Seagate Rescue data recovery service included (2 yr). Orange rubber sleeve — iconic and unmistakable in field kit. Far more appropriate than a standard drive for emergency deployments.
USB flash drive 32 GB	\$10.00	E	amazon.com	~\$8–12 street	Sneakernet file transfer, installer media. Commodity.
USB GPS receiver	\$35.00	E	amazon.com	~\$30–40 street	u-blox chipset (VK-162 or BU-353S4). Time + position for dashboard, APRS. Commodity.

VERIFIED SYSTEM TOTALS — July 3, 2026

FieldComms Server (Pi 5 16GB, Pironman MAX, 2× 1TB NVMe RAID, PSU, SD)	\$784
44Net Gateway (Pi 5 16GB, Argon NEO 5, 256GB SATA SSD, PSU)	\$399
Networking (3× ASUS RT-BE58 Go @ \$119, UniFi Lite 16 PoE @ \$199, cables)	\$586
Cellular WAN (InstyConnect Drum @ \$375, Switchblade @ \$425, cable @ \$20)	\$820
Satellite WAN (Starlink Standard kit @ \$349, adapters)	\$392
Operator Workstations (4× Pi 500+ Desktop Kit @ \$430, 4× Pi Monitor @ \$100)	\$2,120
HF Station (IC-7300MK2 @ \$1,500, 2× Astron RS-35M-AP @ \$344 ea, LDG IT-100 @ \$180, cable, laptop @ \$1,000)	\$3,378
PACTOR HF Modem (SCS PXdragon DR-9400 @ \$2,099 + IC-7300MK2 cables @ \$60)	\$2,159
VHF/UHF Station (FTM-510DR, antenna, Digirig \$50, TNC4 \$150, cables verified \$55)	\$1,047
Field Antenna System (MPAS 2.0 @ \$660, URT1 @ \$390, 300 ft wire @ \$300, winders @ \$60)	\$1,410
Transport cases (6× Apache 4800 XL @ \$59.99 list — coupons available)	\$360
Printing & Accessories (MFC-L3780CDW @ \$450, LaCie Rugged 1TB @ \$125, flash @ \$10, GPS @ \$35)	\$620
COMPLETE SYSTEM TOTAL — Verified July 3, 2026	\$14,075

Key changes from prior estimates: (1) Pi 500+ replaces Pi 500 — \$430/kit vs \$130/unit; workstations up \$1,120 total. (2) Starlink kit price DROPPED from \$499 to \$349. (3) Pi 500+ includes 256 GB NVMe SSD and mechanical keyboard; Desktop Kit includes PSU and HDMI — no separate purchases needed. (4) Chameleon USA250 25%-off sale runs through ~July 14, 2026 — MPAS 2.0 could be ~\$495 if purchased now. (5) Icom runs periodic instant coupons — check hamradio.com/special_offers before purchasing MK2. **Recurring costs:** InstyConnect \$5/mo standby / \$79–99/mo active · Starlink ~\$120/mo (or Roam plan, pausable).